



Who's at risk on the road? Demonstrating young and novice driver risk in low-to middle-income contexts

Luis A. Guzman^{a,*}, Ignacio Sarmiento-Barbieri^b, Olga L. Sarmiento^c,
Darío Hidalgo^d, Alex Quistberg^e

^a Grupo de Sostenibilidad Urbana y Regional, SUR, Departamento de Ingeniería Civil y Ambiental, Universidad de los Andes, Colombia

^b Facultad de Economía, Universidad de los Andes, Bogotá, Colombia

^c Facultad de Medicina, Universidad de los Andes, Bogotá, Colombia

^d Departamento de Ingeniería Industrial, Pontificia Universidad Javeriana, Bogotá, Colombia

^e Department of Environmental & Occupational Health, Dornsife School of Public Health, Drexel University, Colombia

ARTICLE INFO

Keywords:

Novice young drivers
Crash risks
Machine learning
Random forests
Logistic Regression
Low- and middle-income countries
Road safety
Colombia

ABSTRACT

Introduction: Young novice drivers experience higher crash rates, yet most studies focus on high-income countries, leaving limited evidence from low- and middle-income countries (LMICs). This study examined crash involvement and its risk factors in Colombia to inform improvements to the national driver-licensing process.

Methods: We analyzed the national driver-license registry from 2007 to 2020 ($n = 5,822,842$) and all police-reported road crashes during the same period ($n = 541,134$). Crash probability was modeled with a non-parametric machine-learning approach (random forests) and, for interpretation, a logistic regression that incorporated age, driving experience, license category, sex, number of fines, and region.

Results: Main effects explained 51% of the predictive variability, while interactions accounted for the remaining 49%. The random-forest model achieved an F1 score of 96.38% with 93.9% of precision, indicating a low false-positive rate and a recall of 98.9%. Driving experience exhibited the strongest interaction effects: interactions with other variables explained $\sim 2.0\%$ of the variability, the pairwise interaction between experience and license category accounted for $\sim 0.4\%$, and the interaction between experience and region explained $\sim 0.25\%$.

Conclusions: Logistic-regression results corroborated the machine-learning findings, revealing negative associations of both age and experience with crash probability across multiple model specifications. As expected, younger and less-experienced drivers faced the highest crash risk. Risk also varied by sex (higher in males), by license category (higher among motorcyclists), and across regions. These findings suggest that implementing a graduated licensing system could reduce crash risk among novice and young drivers in Colombia and, by extension, in other LMICs.

1. Introduction

Motor vehicle crashes (MVCs) represent a persistent and unequal global public health burden. The World Health Organization (WHO) ranks traffic injuries among the leading causes of death worldwide and estimates that 1.19 million people died in 2021, with LMICs accounting for 92% of all road fatalities (WHO, 2023). The burden falls disproportionately on adolescents and young adults, for whom MVCs constitute a leading cause of mortality (Gicquel et al., 2017; Simons-Morton

et al., 2020). Within this age band, risk is sharply stratified. Per-km crash rates among 16-year-olds are roughly double those for 18–19-year-olds and vastly higher than those of drivers aged 30–59 years (Cassarino & Murphy, 2018). Taken together, these findings underscore that the earliest years of independent driving form a critical window for prevention.

The literature indicates that two interrelated mechanisms drive this risk profile: age-related developmental factors and a distinct, experience-related learning curve. Age-based explanations point to

* Corresponding author.

E-mail addresses: la.guzman@uniandes.edu.co (L.A. Guzman), i.sarmiento@uniandes.edu.co (I. Sarmiento-Barbieri), osarmien@uniandes.edu.co (O.L. Sarmiento), dariohidalgo@javeriana.edu.co (D. Hidalgo), daq26@drexel.edu (A. Quistberg).

<https://doi.org/10.1016/j.jsr.2026.03.015>

Received 19 September 2025; Received in revised form 6 February 2026; Accepted 31 March 2026

Available online 15 April 2026

0022-4375/© 2026 The Author(s). Published by Elsevier Ltd. This is an open access article under the CC BY-NC license (<http://creativecommons.org/licenses/by-nc/4.0/>).

cognitive and psychosocial maturation. Young drivers are more likely to exhibit impulsivity, overconfidence, sensitivity to immediate rewards, and vulnerability to peer influence, which are associated with speeding, tailgating, and unsafe passing (Gheorghiu et al., 2015; Scott-Parker & Weston, 2017; Senserrick et al., 2017). Cross-national evidence consistently highlights elevated crash involvement for drivers aged about 16–25 in the United States, Canada, and the European Union, as well as in Australia and New Zealand (CDC, 2022; Oviedo-Trespalacios & Scott-Parker, 2018; Senserrick et al., 2018; Williams et al., 2012). The United Kingdom and France report similar patterns of disproportionate crash involvement among drivers aged 17–24 and 15–24, respectively (Gicquel et al., 2017). These shared trends suggest a robust association between young age and crash risk across high-income settings.

The literature also cautions, however, that chronological age alone does not fully explain crash risk. Driving experience emerges as an independent and powerful determinant of crashes, even after controlling for age (McCartt et al., 2009; Senserrick & Williams, 2015). Driving is an inherently complex task that requires the simultaneous integration of motor skills, hazard perception, and situational judgment. Although novice drivers may rapidly acquire basic vehicle-control skills, the capacity to perform safely across a wide range of high-risk and dynamic contexts typically develops only through extensive practice (Barua et al., 2014; Keating & Halpernfeldsher, 2008). Inexperienced drivers frequently exhibit diminished hazard perception and weaker vehicle-handling abilities, with these deficits becoming particularly consequential under nighttime or otherwise complex driving conditions (Buckley et al., 2014). Importantly, drivers of any age can be novices: older beginners also face elevated crash risk during the early stages of their driving careers (McCartt et al., 2009). Taken together, these findings indicate that crash risk is best conceptualized as a joint function of age-related maturity and experience-based learning.

These insights have motivated policy interventions that address both dimensions simultaneously. Graduated driver licensing (GDL) is widely regarded as one of the most effective strategies for reducing crash risk among young and novice drivers (Hedlund, 2007; Russell et al., 2011). By increasing supervised driving, delaying exposure to high-risk situations, restricting nighttime driving and same-age passengers, and enforcing zero-tolerance alcohol rules during early licensure, GDL aims to create a safer environment in which experience can be gained gradually (Goldstick et al., 2019; Kaafarani et al., 2015). Since its introduction in New Zealand in 1987, GDL has been widely adopted in Australia, North America, and parts of Europe, and has also been implemented in settings such as South Africa, Hong Kong, and Oman, with many evaluations reporting meaningful reductions in crash and mortality rates (Bates et al., 2014; Boets et al., 2016). Conversely, recent evidence suggests that some stand-alone training and educational interventions may be insufficient to reduce MVC risk among young and novice drivers, reinforcing the case for policies that manage exposure rather than relying solely on knowledge-based approaches (Hafetz et al., 2025).

Despite this robust global evidence base, research on novice and young drivers in low- and middle-income countries (LMICs) remains comparatively limited. Published work in LMIC settings often focuses on motorcyclists, risky behaviors, or evaluations of isolated training programs (Gifty et al., 2021; Matheka et al., 2015; Soori, 2014; Sumit et al., 2021). However, systematic analyses that disentangle and jointly model the impacts of age and driving experience—while also accounting for gender, vehicle type, and geography—are still scarce. In Colombia, existing studies largely describe overall road death counts, relate trends to enforcement or fleet expansion (particularly motorcycles), and benchmark outcomes against other countries (Cabrera et al., 2016). These contributions confirm the high prevalence of road deaths among young adults but leave unresolved the specific mechanisms through which age and inexperience combine to produce heightened risk.

This study directly addresses that gap by examining Colombia as a critical LMIC case in which motorization is expanding rapidly, primarily

on motorcycles, while institutional data and enforcement constraints persist. Using national databases on driving licenses (8.5 million records), road crashes (1.9 million records), and traffic fines (9.3 million records) for 2007 to 2020, this study implements a two-stage modeling strategy. First, random forests are used to detect complex and potentially nonlinear interactions among age, experience, and other key covariates. Second, logistic regression is employed to estimate these effects within a transparent, policy-oriented parametric framework.

2. Context and case study

Among the countries included in the 2024 International Transport Forum's Road Safety Annual Report, Colombia records the second-highest road traffic fatality rate per capita, with 15.8 deaths per 100,000 inhabitants in 2023, behind Costa Rica (16.8) and above the United States (12.2). Colombia also exhibits the highest fatality rate per registered vehicle, at 4.4 deaths per 10,000 vehicles in 2023, followed by Chile and Costa Rica (both at 3.1). In contrast, the best-performing countries report fewer than 3 road traffic deaths per 100,000 population and fewer than 0.3 deaths per 10,000 registered vehicles. National totals have also risen in recent years: traffic deaths increased from 6,505 in 2017 to 8,433 in 2024, representing a 1.34% decrease over 2023. However, official statistics do not disaggregate fatalities by age group, vehicle type, road geometry, or weather conditions. Even so, Colombia reported a rate of 12.6 deaths per 100,000 inhabitants between 2011 and 2020 (ANSV, 2022). Given this substantial and growing burden, there is an urgent need for evidence-based and effective policies to address this public health issue. Identifying who the main road users involved in MVCs are, and where these incidents most commonly occur, is essential for informing relevant policy decisions.

Fig. 1 presents the rates of deaths and injuries resulting from MVCs in the country during 2023. As mentioned earlier and shown in the maps, Colombia experiences high rates of traffic deaths and injuries, with substantial regional variability. In general, areas where motorcycle use is predominant tend to exhibit higher fatality and injury rates (for instance, in Orinoquia, with flat terrain and a lower-density, lower-quality road network), although no clear trend explains the underlying causes.

To give context to our analysis, in Colombia, licenses are classified into three broad categories: Category A licenses are designated for motorcycles and three-wheelers; Category B for private cars; and Category C for public transport and commercial vehicles.¹ Each category is further subdivided: A1 is for motorcycles with a cylinder capacity of ≤ 125 cc, and A2 covers motorcycles and tricycles with a cylinder capacity greater than 125 cc. B1 allows the operation of cars, SUVs, and minibuses; B2 includes light trucks, vans, and buses. C1 is for public service cars, SUVs, and minibuses (taxis and special services under hire), and C2 applies to trucks, vans, and buses used for public service. Specific requirements apply to obtain each license category; however, some of these requirements may overlap. The minimum legal age to obtain a driver's license is 16, and no graduated licensing system is currently in place. Applicants must complete 90 h of training at an approved driving school (60 h of classroom instruction and 30 h of practical training) and pass both medical and reaction-time tests. Most people obtain their first license between ages 16 and 25, while first-time licensing after age 60 is rare. License categories A and B can be obtained directly at age 16, whereas Category C requires holding a Category B license for at least two years, along with additional training (30 h of theoretical instruction and 15–30 h of practical training).

The reported crash rates involving young drivers may be even more pronounced in countries like Colombia, where enforcement is weaker

¹ Although this license category is not common among young and inexperienced drivers, we included this category in the analysis for informational and comparative purposes.

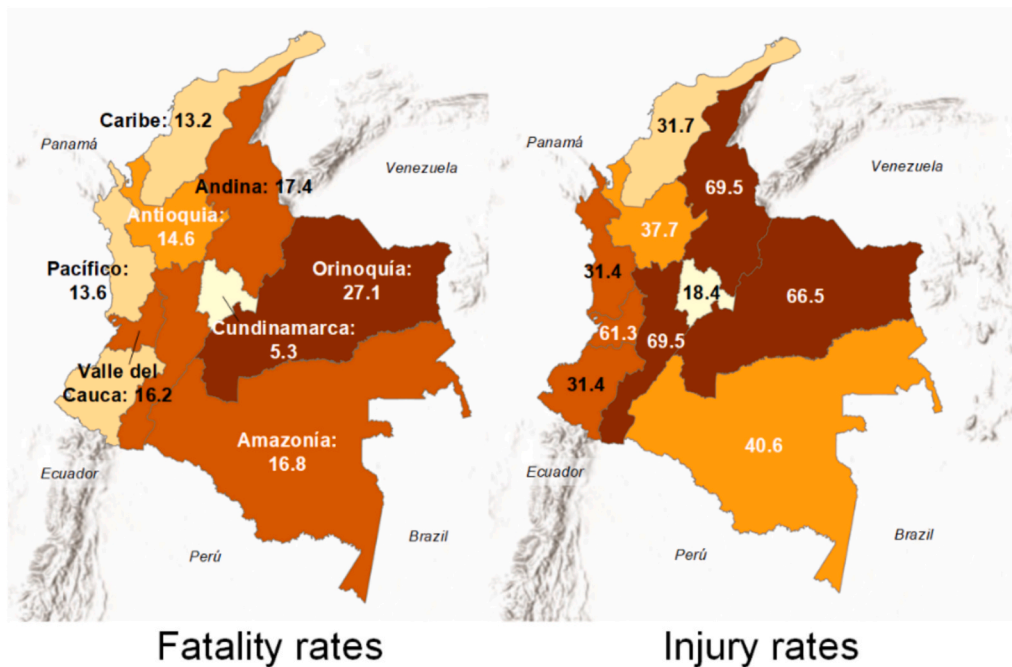


Fig. 1. Fatality and injury rates per 100,000 inhabitants per region, 2023.

than in industrialized nations and where licensing systems are not as robust or thoroughly implemented, and the process to obtain it is not transparent in many places. There is limited evidence regarding the incidence and risk of traffic crashes involving young, novice drivers in developing countries. Using the available information, this study analyzes the interaction between crash characteristics, driver attributes, and vehicle types. The aim is to better understand the patterns of road crashes based on variables such as driver age, experience, and sex, as well as collision-related factors like vehicle type and geographic region.

3. Data and methods

We first describe the data and the construction of key variables. We then outline our two-stage modeling strategy, which integrates random forests and logistic regression to identify and interpret interaction effects among risk factors for MVCs. This approach leverages the predictive accuracy of machine learning while retaining the interpretability of traditional statistical models, thereby providing valuable insights into the factors contributing to crashes and informing policy decisions. It is important to highlight that our assessment was restricted to MVCs and does not reflect information related to other factors, such as traffic composition, roadway geometry, or land use, since this information does not exist.

3.1. Data

Our study draws on comprehensive administrative records covering

driver's date of birth and gender, the issue and expiration dates, and the municipality where the license was issued. Unfortunately, there is no additional information about the drivers or the context that could be used in this type of analysis.

Because Colombia has a large number of municipalities, we aggregate them into the country's natural regions (see Fig. 1), defined by shared economic, demographic, and geographic characteristics. For reporting purposes, however, we merge the three largest cities: Bogotá, Medellín, and Cali, with their corresponding departments (Cundinamarca, Antioquia, and Valle del Cauca, respectively).

We matched these license records with official MVC crash reports, which record each crash's date and severity (injury or fatality). We also integrated data on traffic violations (fines), enabling us to examine how prior infractions influence crash risk. Our primary explanatory variable, driving experience, is defined as the number of years between the license-issuance date and either the date of a crash (for drivers involved in crashes) or December 31, 2020 (for drivers with no crash on record during the observation period).

3.2. Empirical methods

We model the probability that an individual is involved in a crash as a function of several key factors: age, driving experience, number of prior fines, gender, license category, and region. Formally, let $MVC_i = 1$ if driver i was involved in at least one MVC during the observation period (and $MVC_i = 0$ otherwise). We consider a general specification for the crash probability of driver i in Eq. (1):

$$P(MVC_i = 1) = f(\text{Age}_i, \text{Experience}_i, \text{Male}_i, \#Fines_i, \text{LicenseCategory}_i, \text{Region}_i) + \epsilon_i \quad (1)$$

every driving license issued in Colombia between 2007 and 2020. Specifically, we merge three nationwide databases: driving licenses (5.8 million records), road crashes (1.9 million records), and traffic fines (9.3 million records). The license database alone contains 5,822,842 individual entries, each specifying the license category (A, B, or C), the

Where $f(\cdot)$ represents a function determined by our modeling approach. Age_i is the driver's age (in years). Experience_i denotes the number of years the individual has been driving (since license issuance). Besides Experience_i and Age_i , to help adequately profile the risk associated with

Table 1
Random Forest search parameter grid.

Description	Values
Number of trees	100, 300, 600, 1000
Tree depth	2, 3, 4, 5, 6
Minimum number of observations to define a leaf	100, 300, 500

individual drivers, we also add variables associated with driving behavior: the number of fines an individual has received and their gender. Driving ability and the vehicle driven are approximated by the categorical variable *LicenseCategory*_{*i*}, taking levels A, B, and C. Each category corresponds to increasing driving skills and the type of vehicle the driver was likely operating at the time of an MVC. The categorical variable *Region*_{*i*} controls regional variations in driving conditions, such as traffic density, road quality, and traffic law enforcement, that could influence the likelihood of MVCs. Finally, ϵ_i represents an error term capturing unobserved factors.

To uncover the contributions of these factors and their interactions, we use a two-stage modeling strategy that combines a machine-learning model (random forest) with a traditional parametric model (logistic regression). In the first stage, we fit a random forest classifier to identify complex nonlinear relationships and higher-order interaction effects among the predictors of MVC involvement. In the second stage, we estimate a logistic regression that incorporates the main effects and the key interactions identified by the random forest. This approach allows us to compute marginal effects and conduct formal statistical inference. Thus, the random forest serves as a flexible tool for pattern detection, while the logistic regression provides a structured framework for estimation and hypothesis testing.

3.2.1. Random forests for non-linear effects

We first estimate a random forest model using a stratified 30% “building” subsample to explore the functional form of $f(\cdot)$ in Eq. (1) and to identify potential nonlinearities and interaction effects among the predictors (see Table S1 in the Supplementary Materials). This approach follows recent work that employs tree-based methods to recover heterogeneous relationships and conditional average treatment effects (Athey et al., 2019). The remaining 70% of the observations are reserved as the “estimation” sample for the second-stage logistic regression analysis, in accordance with the honest estimation principle proposed by Athey and Imbens (2016). The 30% building sample is selected via stratified random sampling based on MVC involvement and region to ensure adequate representation of both rare crash cases and geographic variation.²

We trained the model with an exhaustive grid search over the hyperparameters listed in Table 1, varying the number of trees (100, 300, 600, and 1,000), maximum tree depth (2–6), and minimum node size (100, 300, and 500). This table lists the random-forest hyperparameter grid for model selection: number of trees (ensemble size), maximum tree depth (controls complexity via allowable splits), and minimum observations per leaf (minimum terminal-node sample size). Because only 9% of drivers in the dataset were involved in a crash, we selected the final model configuration using the F1 score obtained from 10-fold cross-validation. This metric jointly summarizes precision and recall and is therefore more informative than overall accuracy in the presence of class imbalance, particularly when the natural class distribution is preserved to identify structural heterogeneity.

² Stratified random sampling was conducted based on two variables: 1) MVC involvement, to ensure adequate representation of the rare outcome of crashes, and 2) region, to preserve the proportional representation of each geographic area. Descriptive statistics for these samples are presented in Tables S1 and S2 of the Supplementary Material Section, which show that the stratified samples are statistically consistent with the main sample.

Beyond evaluating predictive performance, we apply diagnostic tools to examine the model structure and identify sources of nonlinearity and interaction. We first construct partial dependence plots (Friedman, 2001) for age, driving experience, and the number of fines. We then compute the H^2 statistic proposed by Friedman and Popescu (2008) using the trained random forest. This statistic quantifies the gain in predictive accuracy attributable to each pairwise interaction relative to a main-effects model and enables the ranking of interactions for further analysis. For comparability, each H^2 value is normalized by the total variance of the model’s predictions, such that the contribution of each interaction or predictor is expressed as a percentage.

3.2.2. Logistic regression

In the second stage, we estimate a logistic regression model for the crash indicator MVC using the remaining 70% of the estimation subsample (see Table S2 in the Supplementary Materials). This subsample is distinct from the data used to train the random forest, ensuring that model selection and parameter estimation rely on independent samples (Athey & Imbens, 2016). We specify MVC_i as a function of age, driving experience, gender, prior fines, license category, and region, as well as the nonlinear and interaction terms identified by the random forest. This approach provides interpretable estimates of the direction, magnitude, and statistical significance of each effect, thereby complementing the first-stage machine-learning model with a parametric framework for estimation.

To examine how crash risk varies with driving experience across ages, we compute the marginal effect of experience on the predicted probability of an MVC using the estimated logistic regression model and its nonlinear and interaction terms. The marginal effect is defined as the partial derivative of the predicted crash probability with respect to experience and is evaluated with continuous variables fixed at their sample means and categorical variables set at their reference levels. This procedure demonstrates how the impact of an additional year of driving experience varies by age, license category, and region.

4. Results

Our two-stage modeling framework yields two complementary outputs: 1) random-forest results from the first stage and 2) logistic-regression estimates from the second. The random forest pinpoints the most influential predictive factors and their interactions, whereas the logistic regression quantifies the effect of each factor on crash risk in an easily interpretable form. Taken together, these models capture complex, nonlinear relationships while preserving clarity about the key determinants of MVC involvement. The next sections detail these findings, beginning with the descriptive-statistics results.

4.1. Descriptive statistics

Table 2 summarizes the descriptive statistics for our study population, detailing MVC involvement and driver characteristics both overall and by region. The merged dataset includes 541,134 drivers who were involved in at least one reported MVC between 2007 and 2020, representing approximately 9% of all licensed drivers in the underlying registries.³ Although the overall incidence of reported MVCs is modest, it varies substantially across regions: Antioquia shows the highest share of crash-involved drivers (23%), whereas Amazonía shows the lowest (2%). The mean driver age is 35 years (standard deviation [SD] = 11 years), reflecting a broad age range. Average driving experience—calculated at the end of 2020 or at the time of the first crash—is 6.1 years (SD = 3.5 years), with minimal regional variation. Men constitute 67% of the sample; the proportion is highest in the Caribe

³ We treat each event as independent. A few drivers have more than one MVC, but these are negligible.

Table 2
Descriptive statistics of the sample.

Variable	Country level		Regions															
			Amazonas		Andina		Antioquia		Caribe		Cundinamarca		Orinoquia		Pacífico		Valle del Cauca	
MVC	Mean	% (SD)	Mean	% (SD)	Mean	% (SD)	Mean	% (SD)	Mean	% (SD)	Mean	% (SD)	Mean	% (SD)	Mean	% (SD)	Mean	% (SD)
No	5,264,401	91%	77,250	98%	1,364,456	94%	788,231	77%	584,314	94%	1,319,996	93%	242,658	96%	241,878	95%	645,618	90%
Yes	541,134	9%	1,428	2%	85,341	6%	229,513	23%	39,439	6%	92,774	7%	9,771	4%	12,293	5%	70,575	10%
Age	35	(11)	36	(11)	35	(11)	34	(11)	37	(11)	35	(11)	35	(11)	36	(11)	35	(11)
Experience	6.1	(3.5)	6.3	(3.6)	6.4	(3.5)	5.8	(3.7)	6.3	(3.7)	5.7	(3.3)	6.5	(3.6)	6.1	(3.6)	6.2	(3.8)
Sex																		
Male	3,915,776	67%	54,459	69%	949,999	66%	699,156	69%	472,806	76%	935,745	66%	158,362	63%	186,903	74%	458,346	64%
Female	1,889,759	33%	24,219	31%	499,798	34%	318,588	31%	150,947	24%	477,025	34%	94,067	37%	672,68	26%	257,847	36%
# Fines	0.56	(1.2)	0.43	(0.98)	0.46	(1.0)	0.53	(1.3)	0.72	(1.6)	0.58	(1.1)	0.4	(0.84)	0.56	(1.2)	0.69	(1.4)
License Category																		
A	2,393,449	41%	45,866	58%	659,837	46%	483,815	48%	236,030	38%	395,212	28%	118,659	47%	111,622	44%	342,408	48%
B	2,461,733	42%	22,557	29%	559,640	39%	380,208	37%	301,922	48%	703,135	50%	93,631	37%	109,584	43%	291,056	41%
C	950,353	16%	10,255	13%	230,320	16%	153,721	15%	85,801	14%	314,423	22%	40,139	16%	32,965	13%	82,729	12%
N	5,805,535		78,678		1,449,797		1,017,744		623,753		1,412,770		252,429		254,171		716,193	

region (76%) and lowest in Orinoquia (63%).

During the evaluation period, drivers received an average of 0.56 traffic-violation tickets (SD = 1.2). The distribution is highly skewed: while most drivers incurred no fines, a small number accumulated several. Driver's licenses are classified into Categories A, B, and C, which denote ascending levels of certification. Category B is the most prevalent (42% of licenses), followed closely by Category A (41%) and Category C (16%). The distribution also varies regionally: Category A dominates in Amazonía (58%), whereas Category B is most common in Cundinamarca (50%), indicating regional differences in vehicle use and driving purpose.

4.2. Random forest results

After cross-validation, the best-performing random-forest model used 300 trees, a maximum tree depth of 6, and a minimum leaf size of 100 observations. Table S3 (see in the Supplementary Materials) reports the performance metrics for the four top-ranked configurations we evaluated. The selected model achieved the highest F1 score—96.38%—while the other leading configurations (e.g., models with 100, 600, or 1,000 trees and the same depth) produced nearly identical F1 scores of 96.37–96.38%. This consistency suggests that, once sufficient model complexity is reached, overall performance is robust and not overly sensitive to the exact hyperparameter choices. On the test set, the chosen model attains a precision of 93.93% and a recall of 98.95%. The high precision implies a low false-positive rate: when the model predicts that a driver will be involved in a crash, it is correct about 94% of the time.

In addition, after training the random-forest model on a representative sample of drivers, we examined how it captures the marginal effects of continuous predictors, revealing nonlinear relationships between crash risk and age, driving experience, and number of fines (see Fig. S1). Our analysis indicates that main effects, i.e., the independent contributions of experience, region, license category, fines, gender, and age, explain 51% of the model's total predictive variance. The remaining 49% is attributable to interaction effects, reflecting the extra predictive power obtained when combinations of features are modeled. This decomposition shows that nearly half of the model's predictive performance arises from how variables act in concert rather than in isolation, underscoring the importance of explicitly modeling interactions in crash-risk analysis.

Fig. 2 summarizes the strength of variable interactions, with particular emphasis on experience-related effects. This figure presents Friedman and Popescu's (2008) H^2 interaction statistics for the fitted random-forest model: Fig. 2 (a) presents the overall interaction strength for each feature, defined as the cumulative contribution of that feature's interactions with every other predictor to the model's performance. Each bar represents one predictor, and its height indicates the proportion of explained variability attributable to its interactions. *Experience* exhibits the largest aggregate interaction strength, accounting for roughly 2% of the model's total predictive variance. Fig. 2 (b) isolates individual pairwise interactions. The most influential interaction is between *Experience* and *LicenseCategory*, followed by *Experience* and *Region*. Higher values mean stronger interactions, and all statistics are normalized by the model's total explained variance. These results indicate that the effect of driving experience on MVC risk is heterogeneous and depends on both the vehicle class for which a driver is licensed and the geographic context. Collectively, these findings highlight the multifaceted nature of factors governing MVC risk and underscore the importance of incorporating interaction effects into crash-risk models.

In summary, the random-forest model delivers robust predictive performance and offers valuable insight into the complex interplay among driver characteristics, experience, and crash risk. These findings guide the next stage of analysis, in which we apply logistic regression to quantify these relationships within an interpretable framework.

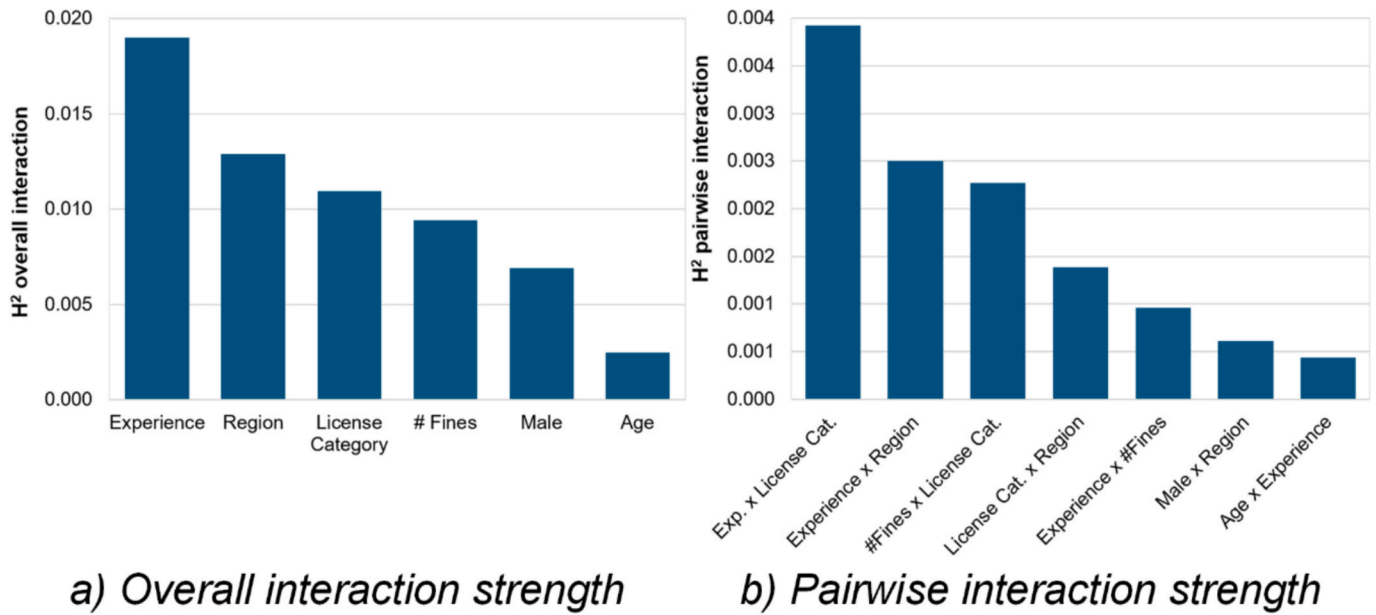


Fig. 2. H² interaction statistics. Note: a) each variable's overall interaction strength (sum of contributions from all interaction terms involving that feature) and b) pairwise interaction strengths for selected variable pairs (the added predictive power from modeling their interaction).

4.3. Logistic regression results

The logistic regression model builds on insights from the random forest, providing interpretable estimates of how each predictor and their key interactions affect crash risk. Table 3 reports three increasingly complex specifications. Model (1) is the baseline and includes only main effects. Model (2) adds nonlinear terms for age and driving experience (quadratic terms and an age-experience interaction) as well as the number of fines. Model (3) incorporates the interaction effects suggested by the partial dependence plots (PDPs) and the H² statistic. By refining the model step by step, we reduce omitted-variable bias and address functional-form concerns, yielding more accurate estimates of the relationships among age, experience, and crash risk. The final specification, shown in Model (3), is formally expressed in Eq. (2).

$$\begin{aligned}
 \Lambda(P(\text{MVC}_i = 1)) = & \beta_0 + \beta_1 \cdot \text{Experience}_i + \beta_2 \cdot \text{Experience}_i^2 + \beta_3 \cdot \text{Age}_i \\
 & + \beta_4 \cdot \text{Age}_i^2 \\
 & + \beta_5 \cdot (\text{Age}_i \times \text{Experience}_i) + \beta_6 \cdot \# \text{Fines}_i + \beta_7 \cdot \# \text{Fines}_i^2 + \beta_8 \cdot \text{Male}_i \\
 & + \sum_{j \neq C} \lambda_j \cdot \text{Experience}_i \cdot \text{LicenseCategory}_{ij} \\
 & + \sum_{k \neq \text{Orinoquia}} \tau_k \cdot \text{Experience}_i \cdot \text{Region}_{ik} \\
 & + \sum_{j \neq C} \gamma_j \cdot \text{LicenseCategory}_{ij} + \sum_{k \neq \text{Orinoquia}} \delta_k \cdot \text{Region}_{ik} \quad (2)
 \end{aligned}$$

In the logistic regression models, we include license category and region as fixed effects to account for time-invariant, unobserved heterogeneity across these groups. These fixed effects absorb differences in MVC risk attributable to structural factors—such as regional driving conditions, traffic patterns, infrastructure, and enforcement policies—and to systematic distinctions among license types, including training requirements and vehicle characteristics. Consequently, the estimated coefficients for experience, age, and their interaction reflect within-group variation rather than between-group differences.

Although the fixed effects are included in the estimation, we do not report their coefficients in Table 3 because they serve primarily as controls for group-specific heterogeneity; their individual estimates are

not substantively informative. Therefore, their estimated values are often of limited interest. Moreover, in nonlinear models such as the logit, fixed-effect coefficients cannot be compared across groups because of the scale dependence of the log-odds metric. Our discussion, therefore, centers on the coefficients of the key explanatory variables, which capture the within-group relationships of principal interest.

Model (1), which excludes interaction terms, shows that age (−0.036) and driving experience (−0.340) are negative and statistically significant predictors of MVC probability. This implies lower crash odds for older and more experienced drivers, consistent with prior evidence. The coefficient on *Male* is positive and significant (+0.853), indicating a higher propensity for crashes among men. By contrast, the coefficient on *#Fines* is negative (−0.360). This counterintuitive result may reflect unobserved heterogeneity, behavioral adaptation (e.g., fines inducing more cautious driving), or the role of fines as a proxy for stricter enforcement environments.

Guided by partial dependence plots and strong H² interactions in Model (2), the specification is extended to include quadratic terms for age, experience, and *#Fines*, as well as an *Age × Experience* interaction. This extension yields a statistically significant improvement in model fit (Table 3). With the quadratic terms, age exhibits an inverted-U pattern: crash risk increases at younger ages, then plateaus and declines at older ages. The negative *Age × Experience* interaction (−0.005) indicates that experience is most protective for younger drivers, while experience remains consistently negative and statistically significant across specifications.

Model (3) is our most comprehensive—and preferred—specification. It adds interactions between experience and license category and between experience and region, following the random-forest results that highlighted these moderators. In this full model, the main effect of *Experience* (interpreted as the effect for the reference category, Category C, and Orinoquia as the reference region) is slightly smaller in magnitude, but two additional interactions stand out: *Experience × Category A* (−0.229) and *Experience × Category B* (−0.055). The first is significantly negative, indicating that each extra year of motorcycle experience (Category A) yields a larger crash-risk reduction than the same year of experience driving commercial vehicles (Category C). The second is negative but not statistically significant, suggesting that the experience effect for car drivers (Category B) does not meaningfully differ from that for Category C drivers. This is very important in LMIC contexts, where

Table 3
Logistic model results.

Variable	Model (1)	Model (2)	Model (3)
Age	−0.036*** (0.003)	0.062*** (0.006)	0.062*** (0.006)
Age squared		−0.001*** (0.000)	−0.001*** (0.000)
Experience	−0.340*** (0.006)	−0.454*** (0.023)	−0.328*** (0.028)
Experience squared		0.025*** (0.000)	0.023*** (0.000)
Experience x Age		−0.005*** (0.000)	−0.005*** (0.000)
Male	0.853*** (0.085)	0.908*** (0.083)	0.922*** (0.086)
#Fines	−0.360*** (0.061)	−0.394*** (0.060)	−0.401*** (0.058)
#Fines squared		0.009** (0.003)	0.010*** (0.003)
Experience x License Category A			−0.229*** (0.038)
Experience x License Category B			−0.055 (0.034)
Experience x Amazonas Region			0.036*** (0.001)
Experience x Andina Region			−0.013*** (0.002)
Experience x Antioquia Region			−0.010** (0.004)
Experience x Caribe Region			0.005 (0.004)
Experience x Cundinamarca Region			0.015*** (0.001)
Experience x Pacífico Region			0.005 (0.004)
Experience x Valle del Cauca Region			−0.005 (0.005)
Obs.	4,063,879	4,063,879	4,063,879
R2 Adj.	0.255	0.269	0.275
R2 Within Adj.	0.198	0.213	0.220
AIC	1,879,198.0	1,843,518.3	1,827,036.6
BIC	1,879,383.0	1,843,756.3	1,827,393.5
Clustered Std. Errors	By region	By region	By region

Note: All models include fixed effects for region and license category. Standard errors are clustered at the regional level to account for intraregional correlation in crash risk. Significance levels: † $p < 0.10$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Robust standard errors are shown in parentheses.

motorcycles are the main individual motorized transport mode (Neki et al., 2023).

Several *Experience* × *Region* interaction terms are also significant, mirroring the random-forest finding that the impact of experience varies

geographically. These differences may reflect local road conditions, the rigor of the licensing process, or regional driving cultures. In some regions, inexperienced drivers may confront particularly hazardous conditions that improve markedly with experience, whereas elsewhere the learning curve is less steep. The sequential introduction of non-linearities and interactions underscores the importance of functional form in MVC risk analysis. The baseline model captures broad trends but oversimplifies relationships by assuming linearity. The model with quadratic terms reveals diminishing marginal effects, and the full model further acknowledges heterogeneity across vehicle types and regions.

Overall, the logistic regression confirms several intuitive patterns: age and experience are protective factors, whereas male gender increases crash odds. It also exposes key nuances: the age–risk relationship is non-linear and intertwined with experience, and the benefit of experience is especially pronounced for motorcyclists and in particular regions. By progressively refining the model specifications, we obtain a more precise and policy-relevant picture of how age, experience, and contextual factors shape MVC risk.

The next section examines these relationships in greater depth by computing the marginal effects of experience across age groups and estimating predicted crash probabilities for representative driver profiles. This allows us to quantify how crash risk evolves with experience and how it varies by age, sex, license category, and region.

4.4. Marginal effects and predicted probabilities

Fig. 3 displays the marginal effects of driving experience on crash risk. The y-axis displays the slope of experience—that is, how the impact of an additional year of driving experience on crash probability changes with age. The x-axis indicates the driver’s age in years. The effect is largest for younger drivers and becomes progressively less negative with age. In practical terms, additional experience dramatically lowers crash risk for young drivers, but the benefit tapers off over time. The steep early decline in crash probability underscores the critical role of initial experience accumulation and supports the use of restrictive licensing measures for novice drivers. By contrast, at older ages, more experience produces little further risk reduction, likely because these drivers have already developed adequate risk-mitigation skills. Overall, the findings highlight the importance of structured early driving experiences in reducing MVC risk among the youngest drivers.

To complement these analyses, we estimate population-averaged marginal effects of driving experience using a marginal standardization approach (Greenland, 2004). Based on the preferred logistic regression specification (Table 3, Model 3), we compute predicted crash probabilities for each driver under their observed level of experience and under a counterfactual scenario in which experience is increased by

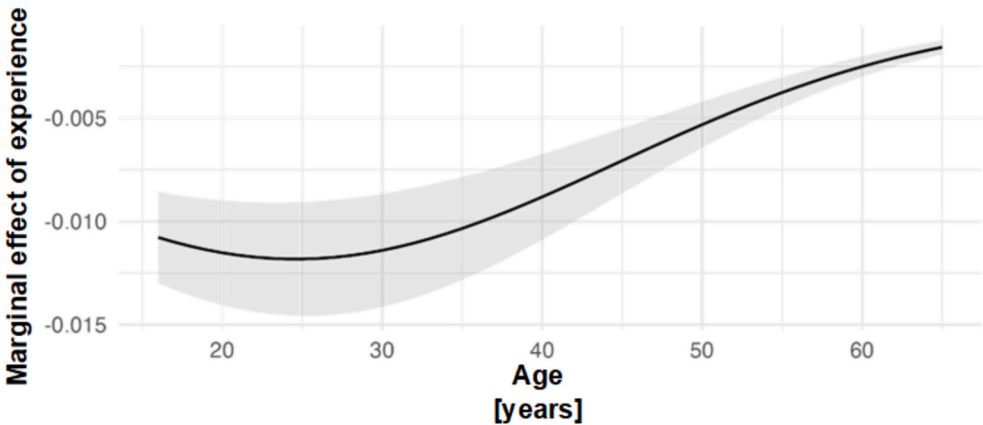


Fig. 3. Marginal effects of experience at different ages. Note: This shows the age-varying marginal effect (slope) of one additional year of driving experience on MVC probability, with a shaded 95% confidence interval; all continuous covariates are held at sample means and categorical covariates at reference levels.

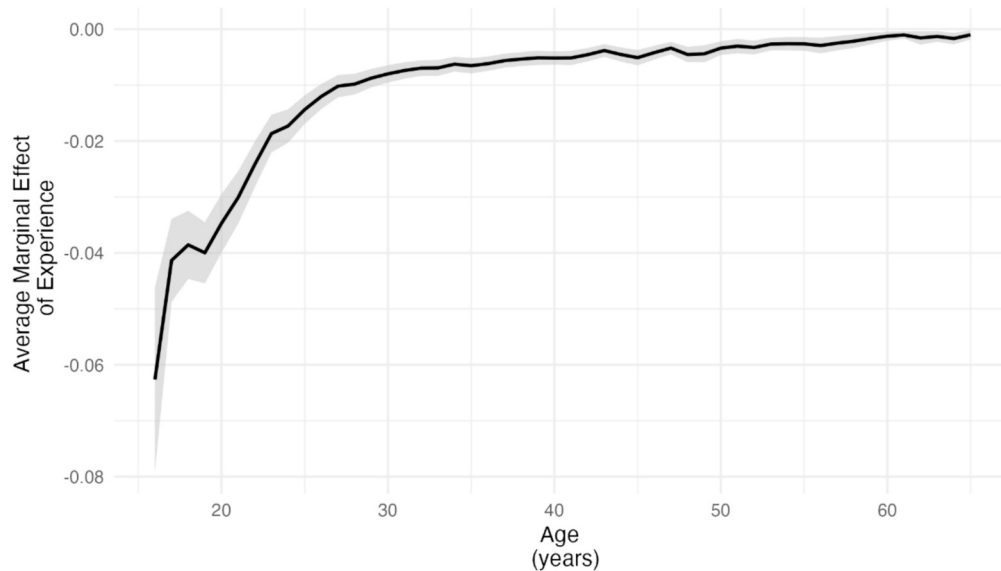


Fig. 4. Average marginal effect of one additional year of driving experience by age. Note: Effects are averaged over the empirical distribution of covariates in the population; shaded areas indicate 95% bootstrap confidence intervals.

one additional year, holding all other observed characteristics constant. Averaging the resulting individual-level differences yields the population-average change in crash probability associated with an additional year of driving experience, standardized to the empirical distribution of driver characteristics in Colombia.

Fig. 4 displays these marginal effects across age groups. The results reveal a pronounced learning gradient: among younger drivers, an additional year of experience is associated with a substantial reduction in crash probability, indicating steep safety gains early in the driving career. As drivers age, the marginal effect of additional experience diminishes and gradually flattens, reflecting decreasing returns once substantial driving exposure has been accumulated.

Because these estimates are averaged over the observed population, they incorporate heterogeneity in vehicle type, gender, enforcement environments, and regional driving conditions. The large reductions in crash risk observed during the early years of licensure underscore the critical importance of managing exposure and supporting skill acquisition among novice drivers, providing direct empirical support for policies such as graduated driver-licensing systems.

Building on the marginal-effects analysis, we calculate the predicted probabilities of an MVC for various driver profiles. These estimates translate the regression results into a more intuitive form, illustrating how crash risk varies jointly with age and driving experience. Fig. 5 plots the predicted crash probability as a function of age (20–65 years) for three representative experience levels: 0 years (novice drivers), 1 year, and 3 years; the surrounding shaded bands denote confidence intervals.

All other covariates remain at their reference categories or mean values to isolate the effect of experience.

The figure shows that drivers with no experience (0 years) have an estimated MVC probability of about 0.35, regardless of age. By contrast, a young driver who has accumulated even a small amount of experience—approximately three years—faces a markedly lower crash probability of 0.13. These values illustrate how quickly MVC risk declines once novice drivers gain practical experience, a pattern that is most pronounced among the youngest motorists, who start at the highest risk.

Driving experience and license category interact strongly in shaping MVC risk. Fig. 5 depicts their joint effect on the predicted MVC probability, revealing distinct risk trajectories that mirror the differing skill demands of each license class. When vehicle type, experience, and sex are considered simultaneously, Fig. 6 provides an even more detailed view of how these variables combine to influence driver safety. This figure shows the predicted probability of an MVC as a function of driver age, stratified by license category (A, B, and C) and sex (female, male). The x-axis displays age in years, and the y-axis displays the predicted probability of a crash.

In Category A, which includes motorcycles, for drivers aged 18 with no experience, the predicted crash probabilities are 0.20 for females and 0.40 for males. After just three years of riding, these probabilities drop sharply to 0.05 and 0.10, respectively, highlighting both the critical role of early experience in motorcycle safety and the gender gap in risk. As riders continue to age and gain experience, crash probabilities decline still further, underscoring the cumulative safety benefits of both factors.

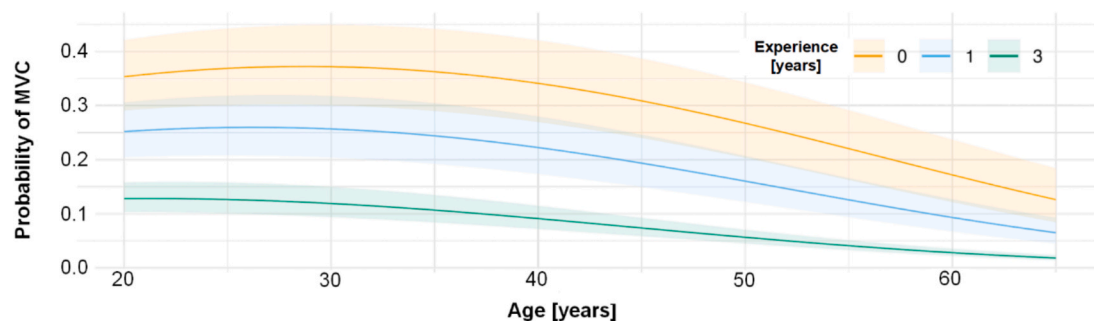


Fig. 5. Predicted probability of crashes by age and experience. Note: Predictions were generated from the logistic regression reported in Table 3, Model (3), with all continuous covariates held at their sample means and categorical covariates set to their reference categories.

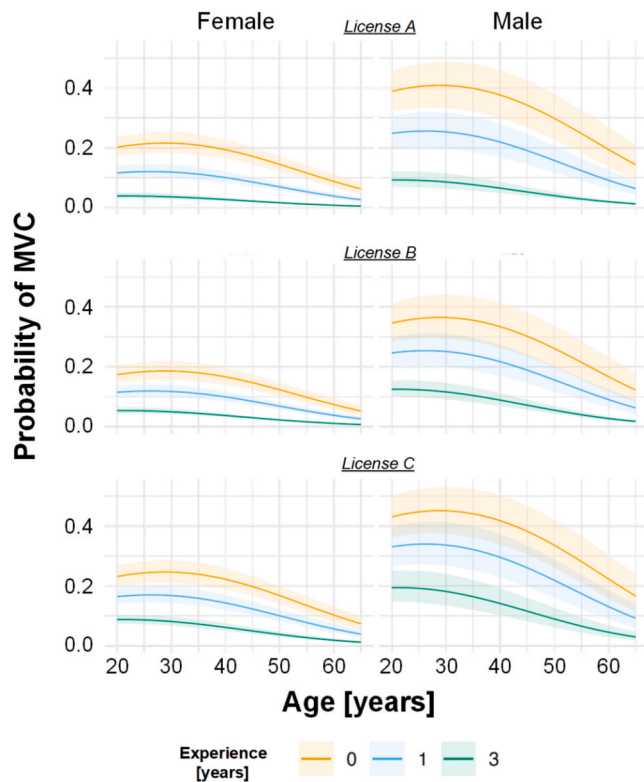


Fig. 6. Predicted probability of crashes across age and experience by gender and category. Note: Each panel shows a license-category/sex combination; lines show 0, 1, and 3 years of experience with 95% CI bands. Probabilities come from Table 3, Model (3) logistic regression, holding continuous covariates at sample means and categorical covariates at reference levels.

In Category B, a similar pattern appears for car drivers. Novices begin with crash probabilities of 0.18 for females and 0.40 for males, which fall to 0.05 and 0.10 after three years of driving. By age 50, these probabilities decline to 0.04 for females and 0.05 for males. Although cars offer a safer baseline than motorcycles, the data confirm that experience remains pivotal in reducing crash risk across all vehicle types.

The bottom panel depicts Category C, which comprises drivers of commercial vehicles that require professional licences. At baseline, the predicted probability of an MVC is substantially higher for novice drivers—0.18 for women and 0.35 for men—underscoring the need for rigorous training and strict licensing standards for operators of large, potentially more hazardous vehicles. In this category, crash probabilities also decline the most sharply as age and experience accumulate, indicating that the combined effects of maturation and practice markedly reduce risk.

Across all categories, a consistent pattern emerges: initially elevated crash probabilities fall with both increasing age and driving experience, and the decline is most pronounced among younger drivers. This finding highlights the safety benefits of graduated and phased driving exposure, particularly for novices. The gender differences observed across categories likewise suggest distinct risk profiles associated with sex and vehicle class.

Beyond individual attributes, the safety gains associated with experience vary geographically. Fig. 7 plots predicted MVC probabilities over age-experience profiles for each region, revealing distinct trajectories. The x-axis shows age (years), and the y-axis shows the estimated probability of a crash; the three plotted lines correspond to drivers with 0, 1, and 3 years of experience; shaded bands denote 95% confidence intervals. In some regions, crash risk drops steeply with additional experience; in others, the effect is more gradual. For example, in

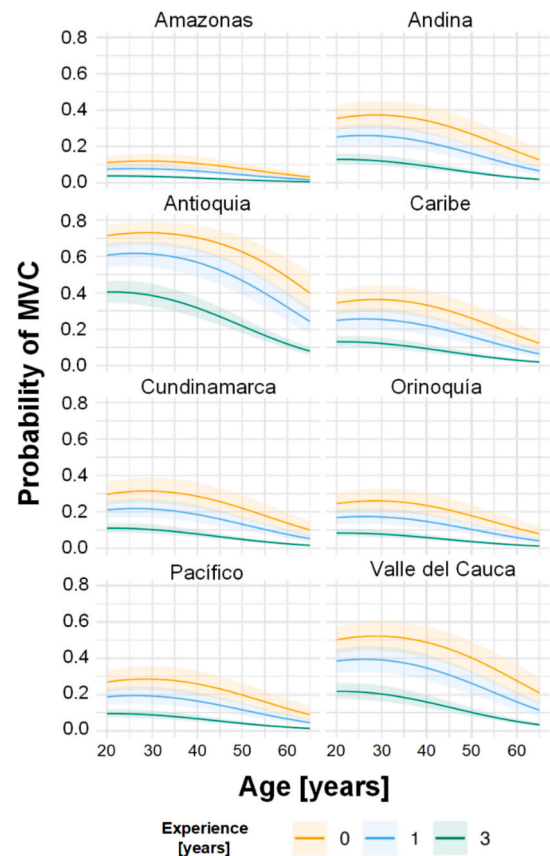


Fig. 7. Predicted probability of crashes across age and experience by region. Note: Predicted probabilities come from the logistic regression in Table 3, Model (3), with continuous covariates fixed at their sample means and categorical variables set to reference categories.

Amazonas, the probability curve remains relatively flat: among 20-year-old drivers with no experience, the estimated crash probability is about 0.10, and it falls only to roughly 0.05 after three years. At age 50, the reduction is similarly modest, implying that while experience still confers safety benefits, its marginal effect may be dampened by lower traffic density or simpler road conditions in that region.

By contrast, in Antioquia, the model yields a pronounced peak in crash probability for inexperienced, middle-aged drivers, followed by marked reductions as experience accumulates. For example, young drivers begin with an estimated MVC probability of 0.70 when they have no experience, which falls to roughly 0.40 after three years behind the wheel. A 50-year-old driver's probability declines from 0.60 with no experience to 0.20 with three years of experience. These results underscore the pivotal role of experience in managing the complex driving conditions characteristic of the region.

Overall, the regional analyses show that while accumulated driving experience generally lowers crash risk, its effect is heterogeneous. The magnitude of risk reduction appears to depend on region-specific factors—such as road infrastructure, traffic density, and enforcement practices—that interact with driver attributes to shape MVC probabilities.

5. Discussion and practical implications

This study examines driver crash involvement and associated risk factors in Colombia, aiming to generate robust evidence to improve the country's licensing process. The analysis combines descriptive statistics, random-forest modeling, logistic regression, marginal-effects estimation, and regional comparisons. Applying this mix of state-of-the-art

crash-analysis techniques in an LMIC context is still uncommon, making our contribution distinctive. Our two-stage framework represents a methodologically innovative approach that strategically integrates the exploratory power of machine learning with the inferential rigor of parametric modeling. This combination of predictive accuracy and interpretability balances exploratory discovery with policy-oriented inference. This integrated methodology is designed to balance predictive discovery with interpretability, an especially valuable combination in settings where data quality and completeness may be uneven.

Beyond its methodological contribution, the study is explicitly oriented toward policy impact. Its findings provide evidence relevant to licensing restrictions and the design of risk-reduction measures for novice young drivers in Colombia. This evidence has already informed a proposed national graduated licensing bill, which has passed two of the four required debates in the Colombian Congress. By foregrounding the interconnected roles of age and experience, the study reinforces the core narrative of the international literature while extending it into an understudied LMIC context. The overarching objective is to clarify how developmental vulnerability and early-career driving inexperience jointly shape crash risk, thereby enabling more precise, evidence-based interventions to protect young drivers and other road users.

The results provide critical and practical insights for shaping policies that reduce MVCs in Colombia and comparable LMICs. We confirm an elevated crash risk among young and novice drivers—especially motorcyclists—and show that driving experience markedly mitigates that risk. Several complementary countermeasures emerge: enhanced education and enforcement; stricter control of speeding, handheld-device use, and driving under the influence of alcohol; and promotion of safer vehicles (Oviedo-Trespalacios & Scott-Parker, 2018). We also recommend introducing a graduated driver-licensing (GDL) system, which has produced demonstrable safety gains elsewhere (Bates et al., 2014; Boets et al., 2016).

These findings are already informing legislative action. A bill now progressing through the Colombian Congress—approved in September 2025 in the second of four required debates—would impose restrictions on novice drivers during their first two years of licensure. Modeled on successful GDL frameworks in high-income countries, the bill proposes nighttime-driving limits, passenger restrictions, and zero-tolerance alcohol policies to foster safer behavior during the riskiest early stage of driving. Although our study could not disaggregate crashes by time of day or weather, nor ascertain whether drivers were accompanied or impaired, these variables have proven vital in GDL programs elsewhere and merit attention in LMIC settings.

To maximize lifesaving impact, policymakers should adopt phased licensing systems that gradually relax restrictions as drivers gain experience, especially for motorcycle license holders (Category A), who face the highest initial risk yet accrue the greatest safety benefits from practice. Strengthened training and testing are equally important for motorcycles and commercial vehicles, where skill deficits disproportionately raise MVC risk. Practical exams should incorporate risk-perception modules covering scenarios such as nighttime travel and passenger distraction, where inexperience and cognitive immaturity intersect (see Section 2). Regional tailoring is also essential: in Antioquia, for example, higher MVC rates among inexperienced middle-aged drivers suggest that targeted infrastructure upgrades (e.g., clearer signage, better road maintenance) combined with localized enforcement could amplify the protective effect of experience.

Enhanced enforcement—particularly tighter alcohol and speed controls—should accompany licensing reforms. Our data indicate that fines correlate with lower MVC risk, presumably by encouraging behavioral adjustments under increased oversight. Emerging technologies such as telematics and dashcams could help monitor compliance with GDL restrictions. Concurrent public-awareness campaigns should involve families and communities, emphasizing parental oversight and addressing gender-related risks, notably the overrepresentation of young male motorcyclists in crash statistics. Cross-sector collaboration

among national agencies, local governments, and insurers can harmonize licensing standards, training curricula, and insurance incentives to reinforce systemic safety measures.

Colombia's legislative initiative offers a replicable roadmap for LMICs grappling with rapid motorization and underdeveloped licensing regimes. The study's focus on motorcycle safety is particularly relevant, given the prevalence of two- and three-wheeled vehicles in many LMICs. Evidence-based measures—such as context-specific GDL schemes—can address the dual challenges of driver inexperience and infrastructural shortcomings. Finally, rigorous impact evaluations, including pilot programs that test extended learner periods, will allow continuous policy refinement. Colombia's experience illustrates how academia, government, and legislators can collaborate to translate research into life-saving action, providing a model for global efforts to create safer roads.

Finally, although the dataset used in this analysis is relatively comprehensive, several limitations merit consideration. First, the use of limited data about the drivers and the road environment. Also, the available information covers only reported MVCs, so some incidents are likely underreported. Crash data reporting has changed over time. Until 2022, crashes could be reported as property-damage-only, injury, or fatal events; however, property-damage-only crashes were heavily underreported. Since 2022, the National Police have officially reported only injury and fatal crashes. Injury counts appear to be especially underreported because official statistics include only cases assessed by the National Institute of Legal Medicine, whereas compulsory insurance records capture a substantially larger number of injuries. Fatality data are more reliable but likely still underestimate true deaths by approximately 5–10% (Rodríguez Hernández et al., 2025). Reflecting broader enforcement and safety challenges, a 2023 report from the National Road Safety Agency indicates that approximately 42% of deceased drivers did not hold a valid driver's license.

Second, necessary data cleaning and record linkage led to the exclusion of some observations. Third, we measured driving experience solely from license-issue dates, which may not perfectly reflect actual years of driving. Finally, the analysis does not account for risk factors such as alcohol use, speeding, weather, time of day, seatbelt non-use, helmet non-use, or vehicle and infrastructure design and maintenance. Factors that could have influenced drivers' behavior, such as road safety campaigns or cultural changes, were also not taken into account during the evaluation period. We have used the best and most detailed information the country currently has. While future research can incorporate these aspects, our findings regarding age and experience remain robust across vehicle categories and regions.

6. Conclusions

This study makes a strong contribution to the limited evidence from low- and middle-income countries (LMICs) on young and novice driver risk by combining a high-coverage national administrative dataset with a two-stage, machine learning-informed inferential strategy. Using random forests to identify nonlinearities and key interaction structures, and then translating those patterns into an interpretable logistic regression framework under an honest-estimation framework (Athey & Imbens, 2016), the analysis illustrates why conventional main-effects-only models are likely to miss substantial heterogeneity in MVC. Using this approach, we find that 49% of predictive variance is attributable to interaction effects, while still maintaining the ability to quantify marginal effects and statistical significance for evidence-based policy design. Although some may view this procedure as unnecessarily complex relative to traditional approaches, the substantial role of interaction effects (49% of predictive variance) underscores that simpler specifications would likely suffer from severe misspecification bias, potentially leading to misguided policy recommendations.

Substantively, the evidence is broadly consistent with patterns observed in high-income settings while revealing LMIC-relevant

nuances. Crash risk is highest among younger and less-experienced drivers, with experience yielding the most pronounced protective benefits in the early years after licensure; this supports the view that the first one to three years of independent driving constitute a critical window for risk management. Risk is also higher among males, higher among motorcyclists, and meaningfully heterogeneous across regions, indicating that the safety returns to accumulating experience depend on both license category and local context. The counterintuitive negative association between fines and crash involvement should be interpreted cautiously as suggestive of deterrence or enforcement-stringency effects rather than as a causal relationship.

These results point to a clear policy package for Colombia and similar LMICs: adopt and tailor a graduated driver licensing (GDL) system that emphasizes phased, supervised, and risk-restricted exposure during the earliest months and years of independent driving, with particular attention to motorcycle license holders. The documented regional heterogeneity also argues for adaptive implementation, pairing national standards with locally calibrated enforcement, improvements in training quality, and targeted infrastructure or safety campaigns in areas where the novice-risk gradient is steepest. More broadly, the study demonstrates how transparent, ML-guided modeling can help bridge the road safety evidence gap in LMICs and provide a defensible empirical foundation for differentiated licensing reforms and future impact evaluations.

CRedit authorship contribution statement

Luis A. Guzman: Writing – original draft, Project administration, Funding acquisition, Formal analysis, Conceptualization. **Ignacio Sarmiento-Barbieri:** Writing – original draft, Investigation, Formal analysis, Data curation. **Olga L. Sarmiento:** Supervision, Funding acquisition. **Darío Hidalgo:** Writing – original draft, Validation. **Alex Quistberg:** Writing – review & editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

We would like to express our gratitude to the National Road Safety Agency (ANSV) of Colombia (www.ansv.gov.co) for the funding and general support given for the development of this study through contract 178-2020. The conclusions and recommendations are solely the responsibility of the authors and do not necessarily reflect the official position of ANSV regarding licensing in Colombia.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jsr.2026.03.015>.

References

- Ansv. (2022). *Estadísticas*. Observatorio Nacional de Seguridad Vial: Comparativo Internacional IRTAD. <https://ansv.gov.co/es/observatorio/estadisticas/comparativo-internacional-irtad>.
- Athey, S., & Imbens, G. (2016). Recursive partitioning for heterogeneous causal effects. *Proceedings of the National Academy of Sciences*, 113(27), 7353–7360. <https://doi.org/10.1073/pnas.1510489113>
- Athey, S., Tibshirani, J., & Wager, S. (2019). Generalized random forests. *Ann. Statist.*, 47(2). <https://doi.org/10.1214/18-AOS1709>
- Barua, S., Sidawi, B., & Hoque, S. (2014). Assessment of the Role of Training and Licensing Systems in Changing the Young Driver's Behavior. *International Journal of Transportation Science and Technology*, 3(1), 63–78. <https://doi.org/10.1260/2046-0430.3.1.63>
- Bates, L. J., Allen, S., Armstrong, K., Watson, B., King, M. J., & Davey, J. (2014). Graduated driver Licensing: An international review. *Sultan Qaboos University Medical Journal*, 14(4), e432. <http://www.ncbi.nlm.nih.gov/pubmed/25364543>.
- Boets, S., Meunier, J.-C., & Kluppels, L. (2016). Implementing graduated driving license in Europe: Literature review on practices and effects, and recommendation of an ideal model. *Rech. Transp. Secur.*, 2016(1–2), 81–96. <https://doi.org/10.4074/S0761898016002089>
- Buckley, L., Chapman, R. L., & Sheehan, M. (2014). Young driver Distraction: State of the evidence and Directions for Behavior Change Programs. *Journal of Adolescent Health*, 54(5), S16–S21. <https://doi.org/10.1016/j.jadohealth.2013.12.021>
- Cabrera, G., Rodríguez, J., & Camelo, F. (2016). Results of the “decade of road safety action” in Colombia. 2010–2015. *Traffic Safety*, A192.3-A193. Doi: 10.1136/injuryprev-2016-042156.534.
- Cassarino, M., & Murphy, G. (2018). Reducing young drivers' crash risk: Are we there yet? an ecological systems-based review of the last decade of research. *Transportation Research Part F: Traffic Psychology and Behaviour*, 56, 54–73. <https://doi.org/10.1016/j.trf.2018.04.003>
- Cdc. (2022). *Leading Causes of death and Injury*. Centers for Disease Control and Prevention.
- Friedman, J. H. (2001). Greedy function approximation: A gradient boosting machine. *Ann. Statist.*, 29(5). <https://doi.org/10.1214/aos/1013203451>
- Friedman, J. H., & Popescu, B. E. (2008). Predictive learning via rule ensembles. *The Annals of Applied Statistics*, 2(3). <https://doi.org/10.1214/07-AOAS148>
- Gheorghiu, A., Delhomme, P., & Felonneau, M. L. (2015). Peer pressure and risk taking in young drivers' speeding behavior. *Transportation Research Part F: Traffic Psychology and Behaviour*, 35, 101–111. <https://doi.org/10.1016/j.trf.2015.10.014>
- Gicquel, L., Ordonneau, P., Blot, E., Toillon, C., Ingrand, P., & Romo, L. (2017). Description of Various Factors contributing to Traffic Accidents in Youth and measures Proposed to Alleviate Recurrence. *Frontiers in Psychiatry*, 8. <https://doi.org/10.3389/fpsy.2017.00094>
- Gift, G., Zubair, S. M., Poobalan, A., & Sumit, K. (2021). Effective interventions in road traffic accidents among the young and novice drivers of low and middle-income countries: A scoping review. *Clinical Epidemiology and Global Health*, 12, Article 100865. <https://doi.org/10.1016/j.cegh.2021.100865>
- Goldstick, J. E., Carter, P. M., Almani, F., Brines, S. J., & Shope, J. T. (2019). Spatial variation in teens' crash rate reduction following the implementation of a graduated driver licensing program in Michigan. *Accident Analysis & Prevention*, 125, 20–28. <https://doi.org/10.1016/j.aap.2019.01.023>
- Greenland, S. (2004). Model-based Estimation of Relative risks and Other Epidemiologic measures in Studies of Common Outcomes and in Case-Control Studies. *American Journal of Epidemiology*, 160(4), 301–305. <https://doi.org/10.1093/aje/kwh221>
- Hafetz, J., Felkins, J., Allston, A., Mann, H., Long, D. L., Ford, C. A., & McDonald, C. C. (2025). The effectiveness of behavioural interventions on young, novice drivers' motor vehicle crash risk: A systematic review. *Journal of Transport & Health*, 43, Article 102045. <https://doi.org/10.1016/j.jth.2025.102045>
- Hedlund, J. (2007). Novice teen driving: GDL and beyond. *Journal of Safety Research*, 38(2), 259–266. <https://doi.org/10.1016/j.jsr.2007.03.003>
- Kaafarani, H. M. A., Lee, J., Cropano, C., Chang, Y., Raybould, T., Klein, E., Gervasini, A., Petrovick, L., DePesa, C., Camargo, C. A., Velmahos, G. C., & Masiakos, P. (2015). The impact and sustainability of the graduated driver licensing program in preventing motor vehicle crashes in Massachusetts. *Journal of Trauma and Acute Care Surgery*, 78(2), 265–271. <https://doi.org/10.1097/TA.0000000000000512>
- Keating, D., & Halpernfeld, B. (2008). Adolescent Drivers: A Developmental Perspective on risk, Proficiency, and Safety. *American Journal of Preventive Medicine*, 35(3), S272–S277. <https://doi.org/10.1016/j.amepre.2008.06.026>
- Matheka, D. M., Faraj, A. O., Kipsaina, C., & Witte, J. (2015). Road traffic injuries in Kenya: A survey of commercial motorcycle drivers. *Pan African Medical Journal*, 21. <https://doi.org/10.11604/pamj.2015.21.17.5646>
- McCart, A. T., Mayhew, D. R., Braitman, K. A., Ferguson, S. A., & Simpson, H. M. (2009). Effects of Age and Experience on Young driver Crashes: Review of recent Literature. *Traffic Injury Prevention*, 10(3), 209–219. <https://doi.org/10.1080/15389580802677807>
- Neki, K., Mitra, S., Wambulwa, W. M., & Job, R. F. S. (2023). Profile of low and middle-income countries with increases versus decreases in road crash fatality population rates and necessity of motorcycle safety. *Journal of Safety Research*, 84, 129–137. <https://doi.org/10.1016/j.jsr.2022.10.014>
- Oviedo-Trespalacios, O., & Scott-Parker, B. (2018). Young drivers and their cars: Safe and sound or the perfect storm? *Accident Analysis & Prevention*, 110, 18–28. <https://doi.org/10.1016/j.aap.2017.09.008>
- Rodríguez Hernández, J. M., Chaparro Narváez, P. E., Hidalgo Troya, A., & Piñeros Garzón, F. S. (2025). Undercounts stemming from misclassification derived from fatal injuries in traffic crashes in Colombia, 2010 to 2021. *Traffic Injury Prevention*, 1–9. <https://doi.org/10.1080/15389588.2025.2495863>
- Russell, K. F., Vandermeer, B., & Hartling, L. (2011). Graduated driver licensing for reducing motor vehicle crashes among young drivers. *Cochrane Database of Systematic Reviews*. <https://doi.org/10.1002/14651858.CD003300.pub3>
- Scott-Parker, B., & Weston, L. (2017). Sensitivity to reward and risky driving, risky decision making, and risky health behaviour: A literature review. *Transportation Research Part F: Traffic Psychology and Behaviour*, 49, 93–109. <https://doi.org/10.1016/j.trf.2017.05.008>
- Senserrick, T., Boufous, S., Olivier, J., & Hatfield, J. (2018). Associations between graduated driver licensing and road trauma reductions in a later licensing age jurisdiction: Queensland, Australia. *PLOS ONE*, 13(9), Article e0204107. <https://doi.org/10.1371/journal.pone.0204107>
- Senserrick, T., McRae, D., & P. P. W., Rome, L. de, Rees, P., & Williamson, A. (2017). Enhancing Higher-Order skills Education and Assessment in a Graduated Motorcycle Licensing System. *Safety*, 3(2), 14. <https://doi.org/10.3390/safety3020014>

- T. Senserrick A. Williams Summary of literature of the effective components of graduated driver licensing systems 2015 https://www.parliament.vic.gov.au/images/Attachment_to_Submission_no._75_-_Literature_Review_of_Effective_Components_of_Graduated_Licensing_Schemes.pdf.
- Simons-Morton, B. G., Gershon, P., O'Brien, F., Gensler, G., Klauer, S. G., Ehsani, J. P., Zhu, C., Gore-Langton, R. E., & Dingus, T. A. (2020). Crash rates over time among younger and older drivers in the SHRP 2 naturalistic driving study. *Journal of Safety Research*, 73, 245–251. <https://doi.org/10.1016/j.jsr.2020.03.001>
- Soori, H. (2014). The Role of Graduated Drivers' Licensing on Incidence and Severity of Road Traffic Injuries in Iran. *British Journal of Medicine and Medical Research*, 4(30), 4918–4925. <https://doi.org/10.9734/BJMMR/2014/8379>
- Sumit, K., Ross, V., Brijis, K., Wets, G., & Ruiter, R. A. C. (2021). Risky motorcycle riding behaviour among young riders in Manipal. *India. BMC Public Health*, 21(1), 1954. <https://doi.org/10.1186/s12889-021-11899-y>
- Who. (2023). *Global status report on road safety*. World Health. Organization. <http://s://www.who.int/publications/i/item/9789240086517>.
- Williams, A., Tefft, B., & Grabowski, J. (2012). Graduated driver Licensing Research, 2010-present. *Journal of Safety Research*, 43(3), 195–203. <https://doi.org/10.1016/j.jsr.2012.07.004>

Luis A. Guzman is an Associate Professor of Civil and Environmental Engineering at Universidad de los Andes. He holds a Ph.D. in Civil Engineering Systems from Universidad Politécnica de Madrid. His research focuses on the economic, social, and spatial impacts of urban transport, including road safety, policy evaluation, land-use interaction, and inequality. He has consulted for the World Bank and the Urban Development Institute of Bogotá and serves on the editorial boards of several transportation journals. His published work covers transport policy, public transport, poverty, and urban structure.

Ignacio Sarmiento-Barbieri is an Associate Professor of Economics at Universidad de los Andes. He holds a PhD in Economics from the University of Illinois at Urbana-Champaign and completed a postdoctoral appointment at the National Center for Supercomputing Applications. His research in urban economics and econometrics integrates AI, machine

learning, big data, and remote sensing to study cities and applied public policy. Recent work covers large-scale correspondence experiments on housing discrimination and climate adaptation projects using satellite imagery. He has published, among others, in *Nature Communications*, *Journal of Public Economics*, *Journal of Urban Economics*, and the *Review of Economics and Statistics*.

Olga L. Sarmiento is a Full Professor in Public Health at the Faculty of Medicine and director of the Epidemiology group at the Universidad de los Andes. She conducts research in the areas of environment and health, including the built environment, physical activity, and obesity in Latin American populations. She is a member of IPEN, the Urban Health Network of Latin America and the Caribbean, and the CRA networks. She directs the physical activity component of the National Survey of the Nutritional Situation of Colombians 2010 and 2015.

Darío Hidalgo is a Notable Professor at the Industrial Engineering Department at Pontificia Universidad Javeriana. He is a civil engineer from Universidad de los Andes in Bogotá, MSc. and PhD. in transportation planning from The Ohio State University. He has spent the last 35 years in urban transport, working both as a government official and as a consultant for various international agencies and local governments in Latin America, Asia, and Africa. He has been a lecturer at Universidad de Los Andes, Universidad del Rosario, and Universidad Externado de Colombia, and is the author of 120+ articles and academic presentations.

Alex Quistberg, PHD, MPH, is Associate Research Professor in the Urban Health Collaborative and the Department of Environmental & Occupational Health. His current research focuses on the epidemiology of road traffic safety, the built environment, and global and urban health. His work implements methods to measure the built environment objectively, including from geographic information systems (GIS), virtual audits, and artificial intelligence via computer vision and deep learning. His work has primarily focused on improving safety for walking and biking, including the evaluation of policies on road safety outcomes.